When Will a Borrower Default?

Survival Analysis in Credit Risk

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Should these be the same outcome?

- One borrower defaults after three months.
- Another defaults after two years.

Both defaulted. The timing is different.

Hypothetical loans.

Principal still outstanding

Illustrative: €12,000 loan; €400 principal repaid each month.

After 3 payments	€10,800 outstanding
After 24 payments	€2,400 outstanding

Earlier default can occur before much principal is repaid.

Assume default just after payment; ignore interest, fees and recoveries.

What do six months tell us?

Observed for six months; no default so far

$$T_D > 6 \text{ months}$$

Provides information for the six months observed.

A binary label needs a horizon

- A classifier can predict default within 12 months.
- Six months without default does not establish a 12-month label.

Training and evaluation labels need appropriate follow-up.

Which 12-month labels are known?

Observed history	Default by month 12?
Default at month 3	Yes
Default at month 24	No
No default through month 6	Unknown

Hypothetical loans; no earlier competing exit in these histories.

Why survival analysis?

- Use when events occur.
- Accommodate right-censored follow-up under suitable assumptions.
- Here: Cox for interpretable relative default hazards.

Machine-learning methods can also model survival outcomes.

Choose the event and the clock

Origination $\longrightarrow$ first recorded default

Elapsed time since the loan was issued

Follow-up can end in different ways

ILLUSTRATIVE · Hypothetical loans; months since origination



C: right censoring — active when observation ends.

D: closure — a competing event ending default exposure.

The survival function

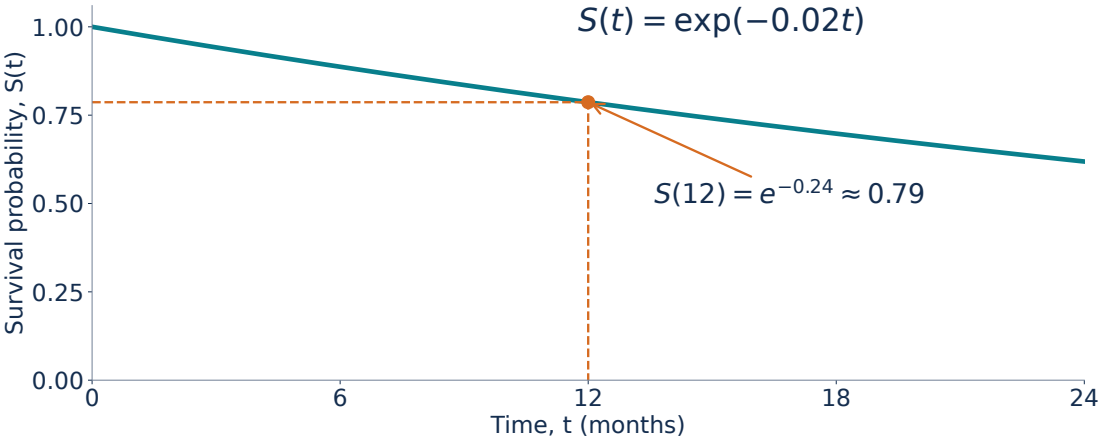
First use a hypothetical default-only setting.

$$S(t) = P(T_D > t)$$

The probability of remaining default-free beyond time t.

Survival: the probability of remaining default-free

Illustrative example · Default-only setting



At 12 months, about 79% remain default-free in this hypothetical setting.

How rapidly are defaults occurring now?

Among loans still exposed now:

how rapidly do defaults occur next?

Hazard: an instantaneous rate, not a probability.

From a short interval to a hazard rate

Hypothetical default-only setting; t in months.

$$h(t) = \lim_{\Delta t \downarrow 0} \frac{P(t \le T_D < t + \Delta t \mid T_D \ge t)}{\Delta t}$$

$$P(t \le T_D < t + \Delta t \mid T_D \ge t) \;\approx\; h(t)\,\Delta t$$

Rate $\times$ a short interval $\approx$ a conditional probability.

From hazard to survival

Default-only setting.

$$H(t) = \int_0^t h(u)\,du$$

$$S(t) = \exp\{-H(t)\}$$

With competing closure, default hazard alone does not determine actual default probabilities.

From timing to borrower characteristics

We have described event timing.

How does the default rate differ between borrowers?

The Cox model

$$h(t \mid x) = \underbrace{h_0(t)}_{\text{baseline}} \underbrace{\exp(\beta^{\mathsf{T}} x)}_{\text{multiplier}}$$

Baseline: the time pattern for reference covariates.

Multiplier: how the covariates scale that rate.

For credit: a cause-specific default rate among loans still exposed.

Compare otherwise identical borrowers

Hypothetical new vs existing customer; same baseline and other covariates.

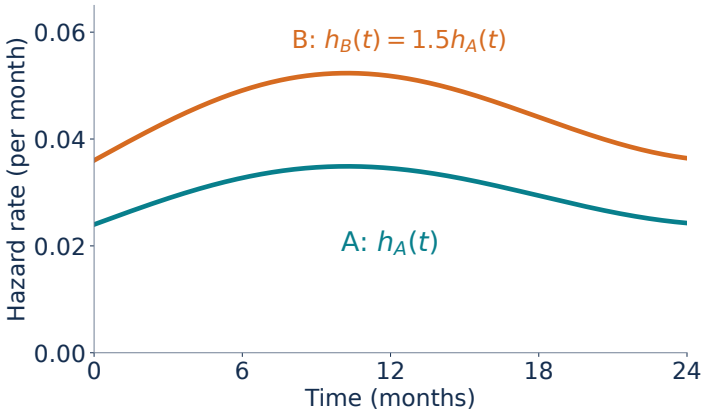
$$\mathrm{HR} = \frac{h(t \mid \text{new})}{h(t \mid \text{existing})} = \exp(\beta_{\text{new}})$$

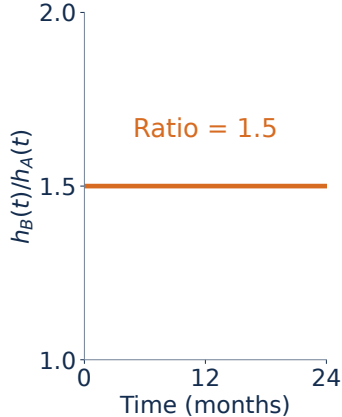
If $\mathrm{HR} = 1.5$: a 50% higher rate at each t.

Proportional hazards: the ratio stays constant over time.

Proportional hazards: changing rates, constant ratio

HYPOTHETICAL · A hazard is an instantaneous rate, not a probability





The baseline hazard can vary over time; proportionality concerns the ratio.

How does Cox learn relative effects?

Illustrative risk set: three borrowers just before one default.

A	B	C
Existing customer	Existing customer	New customer
Relative risk 1	Relative risk 1	Relative risk e^{β}

Which borrower accounts for this default?

Cox regression compares borrowers in the risk set

ILLUSTRATIVE · At risk immediately before a default; all other covariates equal

A	B	C
Existing customer	Existing customer	New customer
Relative risk: 1	Relative risk: 1	Relative risk: e^{β}

$$P(C \text{ defaults} \mid \text{one default in risk set}) = \frac{e^{\beta}}{2 + e^{\beta}}$$

If $\beta = \log(2)$: $\quad \frac{2}{1+1+2} = \frac{1}{2}$

Conditional on one default occurring now; not a probability over a time horizon.

The audited Bondora cohort

Estonia · Issued 2013–2015

Follow-up: at most 730 days from origination

9,844	6,924	2,508
Loans	Borrowers	Recorded defaults

Reporting cutoff: 20 July 2021; all loans have 730 days of potential follow-up.

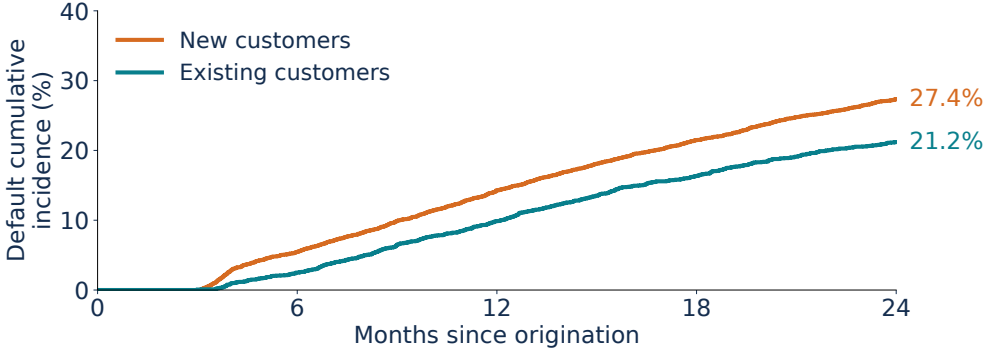
Follow-up uses audited endpoints

- First recorded default date.
- Recorded contract ending for loans labelled repaid.
- Follow-up ends at the first qualifying exit or 730 days.

Closure is a proxy, not guaranteed full repayment.

When do recorded defaults accumulate?

UNADJUSTED / DESCRIPTIVE · Closure proxy treated as a competing event



Loans at risk just before the indicated month

New	6,839	5,971	4,855	3,842	3,081
Existing	3,005	2,683	2,253	1,851	1,534

Aalen–Johansen · No uncertainty bands · Month = 730/24 days · Not a causal comparison

From timing curves to adjusted rates

Timing curves: unadjusted group differences.

Cox: rates adjusted for these characteristics.

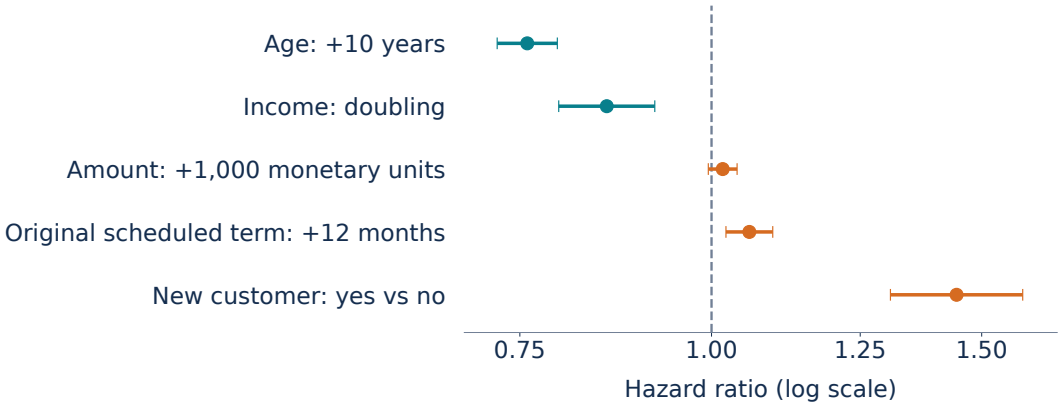
Age	+10 years
Income	Doubling
Amount	+1,000 recorded monetary units
Original scheduled term	+12 months
New-customer status	Yes vs no

Year-stratified baseline; borrower-clustered intervals.

Original monthly-schedule proxy; source currency/income period unverified.

Adjusted associations with the default rate

ADJUSTED · Audited cause-specific Cox model; origination-year baseline strata



Dots: hazard-ratio estimates · Bars: approximate borrower-clustered 95% confidence intervals.

Adjusted associations. Source monetary currency/income period unverified.

New versus existing customer

$$\text{HR} = 1.445$$

95% CI: 1.308–1.595

About 44% higher conditional default rate among loans still exposed.

Association, not a default-probability ratio or causal effect.

Proportional hazards concern: this HR is a summary.

Where interpretation is limited

- Proportional hazards concerns: age and new customers.
- Nonlinearity; approximate tied-date intervals.
- Recorded endpoints and measurement proxies.

Default risk changes over time

- Survival analysis uses both default timing and incomplete follow-up.
- In this cohort, about 27% of new-customer loans and 21% of existing-customer loans defaulted within two years.
- Cox regression connects borrower characteristics to the default rate among loans still exposed.

Group percentages are descriptive. Cox effects are adjusted associations.

Sources

Bondora loan data — snapshot 20 July 2021, via Kaggle.

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Questions?